High-Resolution Remote Sensing Image Segmentation With Global-Guided Normalization and Local Affinity Distillation

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***Abstract*— In recent years, high-resolution (HR) remote sensing images (RSIs) segmentation has received growing attention. The huge number of pixels poses a challenge to the semantic segmentation algorithm, which is limited by the storage of GPUs, so the current methods for processing HR RSIs are categorized into two main categories, i.e., global methods and local methods. The former downsamples the original image and loses a lot of feature details. The latter crops the original image and fails to obtain global contextual information. Both types of methods lead to limited segmentation accuracy. In this article, we propose an end-to-end framework, called global injection network (GINet), which explores two levels of feature distribution and feature relationship to achieve tradeoff between global context and local details. In concrete terms, we propose the global-guided normalization (GGN) module, which injects global context information into local branch and modulates local features using global features to enhance the global perception of local branch. In addition, to constrain the spatial consistency of two branches, inspired by the knowledge distillation technique, we propose local affinity distillation (LAD) loss, which distills the relations in local features into global features to keep the similarity of the relationships corresponding to patches in the two branches. The comprehensive experimental results on three large-scale land-cover classification datasets, DeepGlobe (2448 × 2448), Inria Aerial (5000 × 5000), and GID-15 (7200 × 6800), confirm the effectiveness and superiority of our method in HR semantic segmentation tasks.**

***Index Terms*— Distillation, high-resolution (HR) remote sensing images (RSIs), semantic segmentation.**

I. Introduction

SEMANTIC segmentation is a fundamental task in remote sensing image (RSI) understanding, which aims to annotate each image pixel into different semantic classes of common land-cover objects, such as forests, waters, roads, and so on [1]. It plays an influential role in a wide range of fields, such as autonomous driving, disaster assessment, and environmental monitoring [2]. With advances in opto-electronic devices and remote sensing technology, submeter high-resolution (HR) satellite imagery is becoming increasingly available, providing a wealth of spatial detail and structural information. In the text of [3], an HR image is one that contains millions or even billions of pixels. Typically, a raw, unprocessed Gaofen-2 satellite image has a size of 7200 × 6800, and such HR, ultrawide images present new challenges to semantic segmentation algorithms.

In recent years, convolutional neural networks have injected fresh vigor into semantic segmentation of RSIs. The current advanced general semantic segmentation techniques, FCN [4], DeepLab [5], Unet [6], and so on, cannot directly process HR images due to the limitation of GPU storage. As shown in Fig. 1, one way to process HR images is to downsample them, but this yields low-resolution segmentation results, and after upsampling to the original size, the segmentation accuracy of fine objects cannot be guaranteed, and significant jaggedness will appear in the results. Another approach is to divide it into small patches, use a lightweight network to process each patch individually, and finally splice the results. However, this method does not consider global information, and the prediction results of the same object in the result will be ambiguous.

Fusing global and local segmentation results is an intuitive way to improve the limitations of the above two methods. On the one hand, the global view can provide the global receptive field to solve the field-of-view limitation problem of the local view, and on the other hand, the local view can provide the detail information to compensate for the information loss during the global-view downsampling. Representative works include GLNet [3] and FCtL [7]. However, the GLNet method is a multistage training method, where the global view is first used to train the global network and then used to initialize the weights of the local network, which is too cumbersome. FCtL is a similar, multistage approach that is time-consuming by training multiple models with different context windows and integrating the models to ultimately improve the model's use of the context. Some other methods, represented by ISDNet [8], use a shallow subnetwork to ensure the details of the original image and input the subsampled image into the deep subnetwork to obtain the high-level semantic information, which belongs to the end-to-end hybrid-view method, rather than multistage mixing. It focuses on inference speed and is based on lightweight net-

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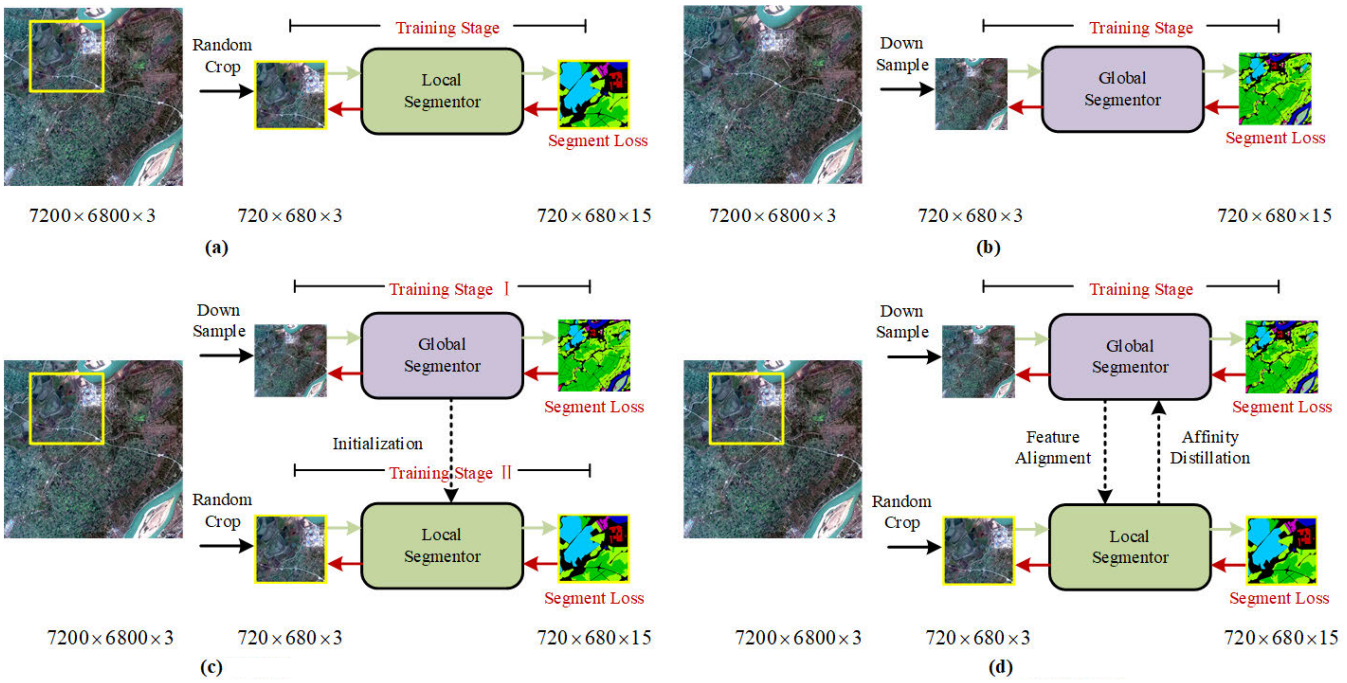


Fig. 1. Comparison of HR image segmentation schemes. (a) Local-view method: the original image is cropped to a small patches, and the final result is obtained by splicing. (b) Global-view method: the original image is downsampled, and the final result is obtained by upsampling. (c) Hybrid-view method: initialize the local model with the global model weights. (d) Our method inputs randomly cropped images and downsampled images, aligns the features by modulation method, and affinity distillation method to align the space.

works, while guaranteeing segmentation performance through novel modules. But, it cannot handle images of large size, and as the image size keeps increasing, it will encounter out of memory, and the algorithm cannot work. The central challenge of the UHR semantic segmentation task is the tradeoff between global context information and local detail information. The sliding-window method can obtain complete details, but it is limited by the size of the window and lacks the control of the global context. The subsampling method can obtain a perfect global view while taking into account the processing speed, but a lot of details are lost in the process of subsampling.

In this article, we propose a global information injection network (GINet) for semantic segmentation of HR RSIs. To fully implement tradeoff between global contexts and local details, we improve the GLNet approach, specifically, introducing a new single-stage framework that utilizes both global and local views for the prediction of results. The core of our framework is the global-guided normalization (GGN) module and the local affinity distillation (LAD) loss. The GGN tunes the features of the local branch by learning biased terms from the global features to give a broader contextual information. In addition, the LAD loss module extracts the relationship matrices of the local patches and the relationship matrices of the corresponding images in the global downsampled images, constraining their consistency. The integration of the GGN module and LAD loss into the GINet model allows better optimization of local and global features and can better handle the segmentation task of HR RSIs.

In summary, the contributions of this article are summarized as follows.

1) We propose a new single-stage end-to-end framework for handling HR RSI segmentation tasks.
2) To enhance the global perceptual capability of the local branch, we develop a GGN module that uses global features to modulate the distribution of local features to achieve a broader object context perception.
3) In order to enhance the detail capturing ability of the global branch, we design an LAD loss to maintain the consistency of the local relationships of global features with the global relationships of local features.
4) We validate the proposed GINet on three HR datasets with different characteristics, DeepGlobe [9], Inria Aerial [10], and GID-15 [11] datasets. Our research shows that GINet can be injected with global information to achieve competitive performance.

The rest of this article is organized as follows. Section II presents the literature work related to semantic segmentation of ordinary resolution and HR RSIs. In Section III, the architecture of GINet and its components are described in detail. Section IV describes the designed experiments and the corresponding experimental results. Finally, we draw the conclusion of this study in Section V.

II. Related Work

A. Semantic Segmentation of RSIs

In the past few years, deep learning methods with FCN [4] as the underlying architecture have been broadly used in the semantic segmentation of remotely sensed images and have demonstrated strong performance. Researchers have made targeted improvements to the structure of the FCN network,

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which can be mainly summarized as a multiscale approach and contextual approach.

In RSIs, semantic segmentation algorithms need to cope with large variations in object scale. Some articles adopted a divide-and-conquer strategy, where different branches deal with objects at different scales. MANet [12] modeled the correlation between different scales to learn multiscale feature representations. SRJDN [13] considered both scale decoupling and relation decoupling to mine out the relationships between multiscale remote sensing objects. Some articles looked at obtaining better multiscale feature representations. AFNet [14] proposed adaptive fusion networks that adaptively focus on objects with size variations on multilevel feature mappings. Peng et al. [15] proposed cross fusion net for fast and efficient extraction of multiscale semantic information, especially for small-scale objects, such as vehicles. Liu et al. [16] proposed ScasNet aggregate multiscale contextual information in a self-cascading manner. Small-scale objects will have their features annihilated after multiple downsampling of the backbone network, so some articles [17], [18] directly adopted an HR backbone network, such as HRNet [19], to keep the features of small-scale objects.

Contextual information can capture long-range spatial dependencies and is widely used in semantic segmentation models for RSIs. A direct way to obtain better context was to increase the receptive field of the network. Dilated convolution operations are widely used. For example, Liu et al. [20] effectively utilized a rich combination of dilation convolution to expand the receptive field of the network with fewer parameters and features. Xiao et al. [21] proposed large field convolution to achieve a large receptive field with fewer parameters. Inspired by the natural language processing, the technique based on self-attention [22] has made a big splash in the field of computer vision, which also yielded better contextual information [23]. Articles, such as [24], [25], [26], [27], and [28], either separately or mixedly, apply the self-attention to the contextual modeling of spatial, channel, category, and region relations to capture the global correlation of features and achieve better segmentation performance. In recent years, the self-attention technique was extended to Transformer-based methods. Zhang et al. [18], Wang et al. [29], and Deng et al. [30] fully utilized the Transformer to get global context information and improved segmentation performance. Zhang et al. [31], He et al. [32], and Xiao et al. [33] combined the local feature representation ability of CNN and the excellent global information perception ability of Transformer to further enhance the performance of the RSI segmentation model. GLOTS [34] used a global–local Transformer decoder to decode coarse and fine features, which provided the segmentation model with global semantic information and local detail information to improve the segmentation performance. However, none of these segmentation methods are specifically designed for HR images.

B. HR Image Segmentation

1) Global-View Method: Downsamples HR images to a suitable scale before feeding them into the network, and then, upsamples the results into an HR semantic mask. MagNet [35] downsampled the global image sequentially into multiple small-scale sequences and bridged the huge gap in the scale space through this progressive refinement operation. ISDNet [8] integrated shallow and deep networks and proposed a relationship-aware feature fusion module to mine the relationship between shallow and deep features. Jin et al. [36] showed that for large images, learning a spatially varying downsampling strategy yields better segmentation accuracy than traditional uniform downsampling. As the image size grows, there still exists the problem of rapid growth in GPU memory usage.

2) Local-View Method: Divides the HR image into patches, obtains the local segmentation results through a lightweight network, and then merges them into an HR semantic mask. FCtL [7] introduced a segmentation model based on local perceptual contextual correlation to deal with local patches. Mfvnet [37] pyramidally sampled the image to construct multiple fields of view, and then, the features of different scales of the images in the multiple fields of view were deep fusion. However, these methods lacked the perception of the global field of view, the accuracy of category determination is limited, and the same object located in different patches will have a category ambiguity due to spatial fragmentation.

3) Hybrid-View Method: Preprocesses HR images into both downsampled images and localized image blocks, and the model combines the segmentation results of both into an HR semantic mask. GLNet [3] integrated the context information of the global branch and the detail information of the local branch, which achieved a tradeoff between memory usage and segmentation accuracy. On this basis, PPN [38] embedded a classification subnetwork in the local branch, which can adaptively get the importance of different patches, tilting the computational resources to the more important patches, further improving the segmentation accuracy. WiCoNet [39] proposed a context Transformer to model the long-range correlation, so that the local branch features can obtain a wider range of contextual information. GeoAgent [40] performed adaptive context scaling operations with the help of reinforcement learning techniques to provided optimal contextual information for different patches and achieved excellent segmentation results in large-scale map applications. GeoAgent focused on the improvement of global thumbnails, while this article's method focuses on fusion methods for global thumbnails and local patch features. SILI [41] explored the task of change detection in the presence of image resolution differences by randomly synthesizing images with different resolutions while introducing a coordinate-based representation to encourage the model to learn scale-invariant features. However, these methods are basically multistage processes, which are cumbersome.

III. Method

We propose a novel framework to solve the task of semantic segmentation of HR RSIs. As shown in Fig. 2, the downsampled view of the original image is fed into the global branch for extracting global context information, and the randomly cropped view of the original image is fed into the local branch

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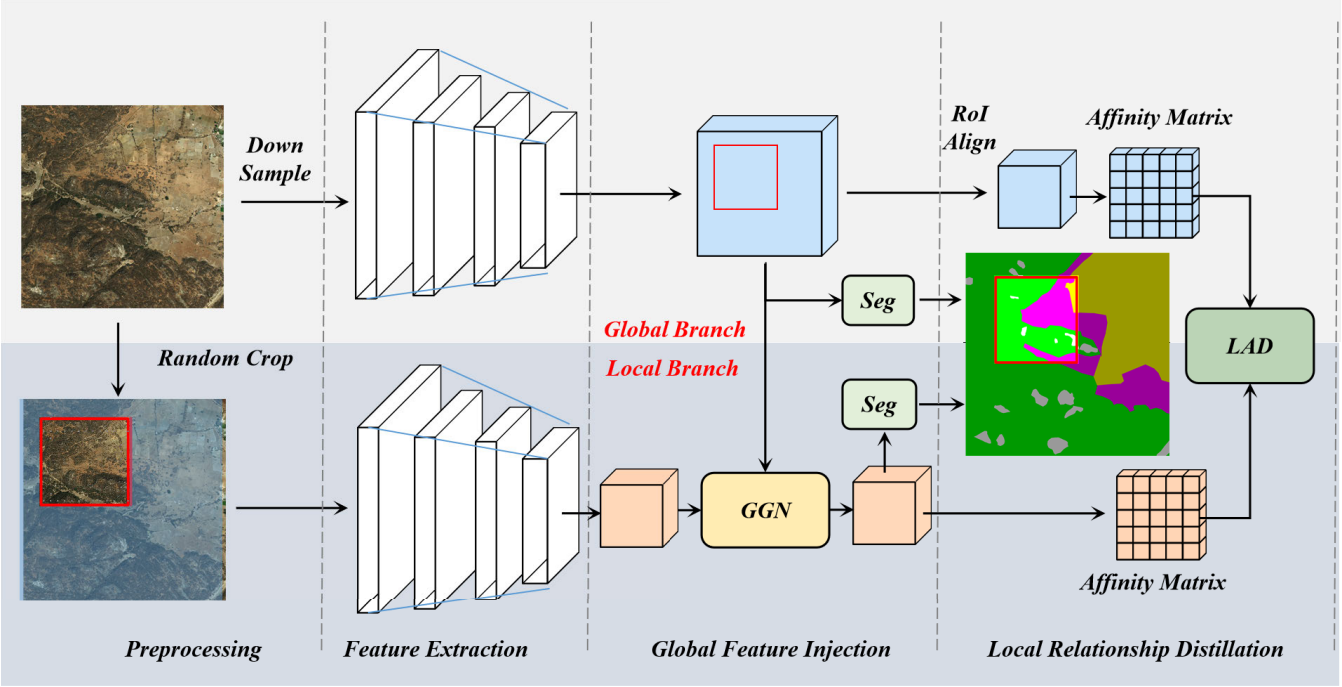


Fig. 2. Proposed GINet process for HR image segmentation. The local features are modulated using global branch features, injecting global information to expand their receptive fields, and using a local relation distillation module to align the spatial locations of the two branches to achieve a consistent characterization of the global and local branches.

for enhancing the spatial details of the segmentation result. In this article, we introduce a GGN module, where the network modulates the local features based on the global semantic features. In addition, we propose LAD loss, which takes the feature relations of local branch to guide the learning of feature relations of global branch.

A. Preliminary

The essence of the task of semantic segmentation is pixel-by-pixel classification, which maps an RGB image $I \in \mathbb{R}^{H\times W\times 3}$ to a semantic map $Y \in \mathbb{R}^{H\times W\times C}$, where H is the height, W is the width, and C is a predefined number of semantic categories. A semantic segmentation model usually consists of a backbone encoder and a segmentation head. The backbone encoder is used to extract image features. The segmentation head decodes the features and produces the output per pixel segmentation. The model is typically trained with the cross-entropy loss.

B. Preprocessing

We adopt a hybrid-view architecture similar to GLNet [3]; i.e., the high-resolution image is subjected to two pre-processing operations, downsampling and random cropping, simultaneously. The specific formalism is defined as follows, downsampling the original high-resolution image I and the corresponding ground truth Y by a factor of s times to obtain the global-view image $I_G \in \mathbb{R}^{h_1\times w_1\times 3}$ and the corresponding ground truth $Y_G \in \mathbb{R}^{h_1\times w_1}$, and feeding the downsampled image into the global branch G. The local branch L receives a cropped patch $I_L \in \mathbb{R}^{h_2\times w_2\times 3}$ and the corresponding ground truth $Y_L \in \mathbb{R}^{h_2\times w_2}$ with the same resolution from I. The model combines the segmentation results of both into an HR semantic mask. Note that I_L and Y_L are randomly cropped into patches during training, while sliding-window cropping is performed during testing, $h_1 = H/s$, $h_2 \ll H$, $w_1 = W/s$, and $w_2 \ll W$.

C. Global Feature Injection

1) Revisit Batch Normalization: We begin by reviewing the formula for batch normalization (BN) [42]. In modern convolutional neural networks, the BN layer is a common operator. It is basically used to adjust and scale activations in neural network, thus accelerating network convergence while solving the covariate drift problem. It consists of two operations: feature scaling and affine transformation. Given an input feature map $X \in \mathbb{R}^{H\times W\times C}$, the process of BN can be formulated as follows:

$$\overline{X} = \frac{X - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}. \tag{1}$$

Equation (1) is a common standardization operation that generates a small batch with a mean of 0 and a unit variance of 1. Here, μ_B is the sample mean of the small batch, and σ_B is the sample standard deviation of the small batch

$$\mathrm{BN}(X) = \gamma \odot \overline{X} + \beta. \tag{2}$$

Since unit variance is a subjective choice, we usually introduce the scale parameter γ and the shift parameter β for adaptive learning, which have the same shape. Note that γ and β are learnable parameters that are adaptively adjusted during network optimization.

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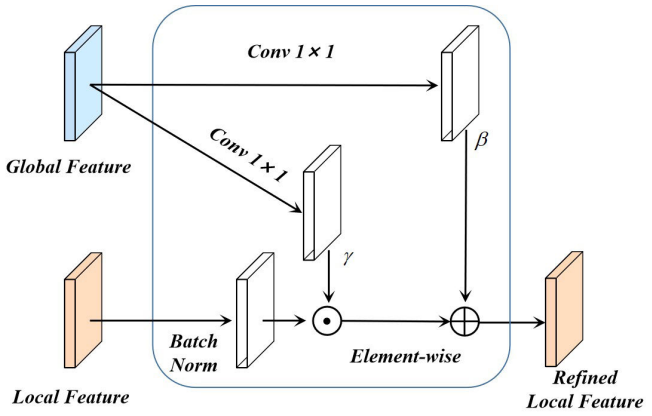


Fig. 3. Proposed GGN module learns the scale parameter γ and the shift parameter β from the global features for modulating the local features.

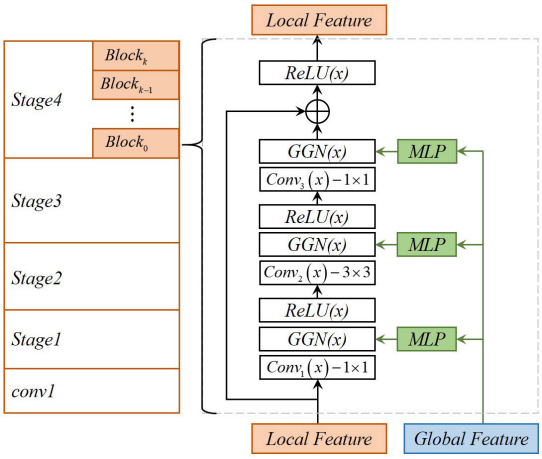


Fig. 4. Proposed Cascade GGN module replaces the GGN module with the original BN layer in stage 4 of the backbone network.

2) Global-Guided Normalization: Inspired by SPADE [43], we apply a similar conditional modulation method to the field of HR image segmentation. In order to expand the receptive field of the local branch, we use global features to modulate the local branch features, as shown in Fig. 3. Specifically, the mean and variance of the local features are first computed; then, the global features are projected into the embedding space, and then, two simple convolutional layers are performed to generate learnable modulation parameters γ and β, respectively. In order to better preserve the spatial information of the local features, distinguishing from with previous conditional normalization methods, we use tensors $\gamma^i_{c,y,x}(F^G)$ and $\beta^i_{c,y,x}(F^G)$ with spatial dimensions. The resulting γ and β are used to adjust the distribution of the local features. The formalization is defined as follows:

$$\overline{F^L} = \frac{F^L - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} \tag{3}$$

$$\mathrm{GGN}\left(F^L, F^G\right) = \gamma^i_{c,y,x}\left(F^G\right) \odot \overline{F^L} + \beta^i_{c,y,x}\left(F^G\right). \tag{4}$$

3) Cascade GGN: Furthermore, in order to improve the detail portrayal ability of local branches and reduce the information loss of the segmentation results caused by excessive downsampling, we refer to [44] to insert the proposed GGN module into the last stage of the local branch backbone network, as shown in Fig. 4, a stage consists of multiple blocks, and we stack three GGN modules in each block; that is, the original BN layers in the block are replaced with our proposed GGN modules. The global features are projected using a layer of MLP before the input module to prevent competition between features.

D. Local Relation Distillation

We propose LAD loss, which aims to achieve consistent characterization of global and local branch. In order to alleviate the spatial misalignment problem of global branching features caused by multiple downsampling in the backbone network, we align the global branching output features and local branching output features in the same region according to the spatial coordinates of the local image blocks. Specifically, in the preprocessing stage, we record the coordinates of the local image block box^L, which are mapped to the corresponding coordinates box^G in the global feature map according to the ratio of the size s

$$\mathrm{box}^G = s \times \mathrm{box}^L. \tag{5}$$

For a patch in the original image, by using the RoIAlign operation [45], we can get the corresponding spatially aligned feature F^G in the global branch. In the local branch, we can get his feature F^L

$$F^G = \mathrm{RoIAlign}(F^G, \mathrm{box}^G). \tag{6}$$

We use the mean square error to define the paired affinity distillation loss

$$\mathcal{L}_{\mathrm{lad}}(F^L, F^G) = \frac{1}{(W' \times H')^2} \sum_{i \in H' \times W'} \sum_{j \in H' \times W'} \left(F^L_{ij} - F^G_{ij}\right)^2 \tag{7}$$

where F^L_{ij} denotes the affinity between the ith pixel and the jth pixel produced by the local F^L and F^G_{ij} denotes the affinity between the ith pixel and the jth pixel produced by the global branch F^G. In our implementation, the similarity between two pixels is simply computed from the features F_i and F_j as follows:

$$F_{ij} = \frac{F_i^{\mathsf{T}} F_j}{\|F_i\|_2 \|F_j\|_2}. \tag{8}$$

E. Overall Loss

We use the traditional pixel-level cross-entropy loss $\mathcal{L}_{\mathrm{ce}}$ to train the local branch and the global branch. In addition, the LAD loss is used to constrain the relationship between the two branches. The overall loss formula is

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{local}} + \mathcal{L}_{\mathrm{global}} + \lambda \mathcal{L}_{\mathrm{lad}} \tag{9}$$

where $\mathcal{L}_{\mathrm{total}}$ is the total loss, $\mathcal{L}_{\mathrm{local}}$ is the loss of local branch, $\mathcal{L}_{\mathrm{global}}$ is the loss of global branch, and λ is the weight of LAD loss, which is used to regulate the relative importance of segmentation and feature alignment tasks in network training.

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TABLE I
THREE HR REMOTE SENSING SEMANTIC SEGMENTATION DATASETS

Dataset	DeepGlobe	Inria Aerial	GID-15
Sample	803	180	150
Class	7	2	15
GSD	0.5m	0.3m	0.8m
Size	2448×2448	5000×5000	7200×6800

IV. EXPERIMENTS

In this section, to evaluate GINet, we conduct experiments on three HR RSI segmentation datasets as shown in Table I, and perform quantitative comparisons with other segmentation methods and ablation study. For fair comparison, the backbone of the generalized segmentation networks is all ResNet-18.

A. Datasets

1) *DeepGlobe Dataset*[1]*:* The dataset focuses on land-cover classification in rural areas and contains 803 HR submeter satellite images of 2448×2448 pixels labeled into seven landscape categories: urban, agriculture, rangeland, forest, water, barren, and unknown, where the "unknown" areas are not considered in the evaluation [9]. We follow the protocol of [3] and divide the images into training, validation, and test sets with 455, 207, and 142 images, respectively.
2) *Inria Aerial Dataset*[2]*:* The dataset is used for building segmentation in aerial imagery and provides 180 aerial orthocolor images of 5000×5000 pixels with a spatial resolution of 0.3 m [10], from five cities, which cover a diverse range of urban settlements, from densely populated areas (e.g., the financial district of San Francisco) to small alpine towns (e.g., Lenze, Tyrol, and Austria), and cover an area of 810 square kilometers (405 km^2 for training and 405 km^2 for testing). We follow the protocol of [3] and divide the images into training, validation, and test sets with 126, 27, and 27 images, respectively.
3) *GID-15 Dataset*[3]*:* This is a large-scale land-cover classification dataset constructed from Gaofen-2 satellite imagery, with each image having 7200×6800 pixels and a GSD of 0.8 m [11]. Specifically, it contains 150 pixel-level labeled images, which are labeled into 15 categories, namely, industrial land (IDL), urban residential (UR), rural residential (RR), traffic land (TL), paddy field (PF), irrigated land (IL), dry cropland (DC), garden plot (GP), arbor woodland (AW), shrub land (SL), natural grassland (NG), artificial grassland (AG), river (RV), lake (LK), and pond (PN). Officially, the dataset is divided into 100 training sets, 10 validation sets, and 40 test sets. Since the test set is not open, this article reports the results on the validation set.

[1]http://deepglobe.org/challenge.html
[2]https://project.inria.fr/aerialimagelabeling/
[3]https://captain-whu.github.io/GID15/

B. Evaluation Metrics

We used the most commonly used metrics to evaluate the test method, including the following: 1) OA, which is the numerical ratio of correctly categorized pixels to all pixels in the image; 2) mean $F1$; and 3) mean intersection and merger ratio (mIoU). These metrics can be calculated in terms of the number of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) pixels, as defined below

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{TN}} \tag{10}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{13}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{14}$$

C. Implementation Details

Both global and local branches use ResNet [46] as the backbone network, and the corresponding pretrained models on ImageNet [47] are used as initialized weights for both branches. Notably, we adapt stage 4 of the local branch using the proposed GGN.

We use the MMSegmentation [48] toolbox based on the PyTorch [49] framework as our codebase and follow the default data augmentation methods to improve the generalization of the models. All model parameters of the segmentation network are optimized using a stochastic gradient descent (SGD) optimizer, with the batch size set to 16, and the momentum and weight decay coefficients set to 0.9 and 5×10^{-4}, respectively; the initial learning rate is set to 0.01, and a "poly" strategy is used to gradually decrease the learning rate. The initial learning rate is set to 0.01, and we use the "poly" strategy to gradually reduce the learning rate, and its power is set to 0.9.

The original image sizes for DeepGlobe, Inria Aerial, and GID-15 are 2448×2448, 5000×5000, and 7200×6800, respectively. For inputs to the global branch, the sizes are set to 1224×1224, 1250×1250, and 1800×1700, and for the inputs of the local branch, the sizes are all set to 512×512. The corresponding maximum number of iterations of the three datasets is set to 20k, 10k, and 10k, respectively.

D. Ablation Study

To further validate our contribution, we perform ablation experiments on the DeepGlobe dataset.

1) Effectiveness of Each Module: In the GINet model, GGN is proposed to modulate the feature distribution of local branch separately, and LAD loss is proposed to maintain the relationship consistency of the same patch in global and local branch. As shown in Table II, experiments with different configurations are conducted to verify the effectiveness of these two modules. The baseline (No. 3) is a segmentation

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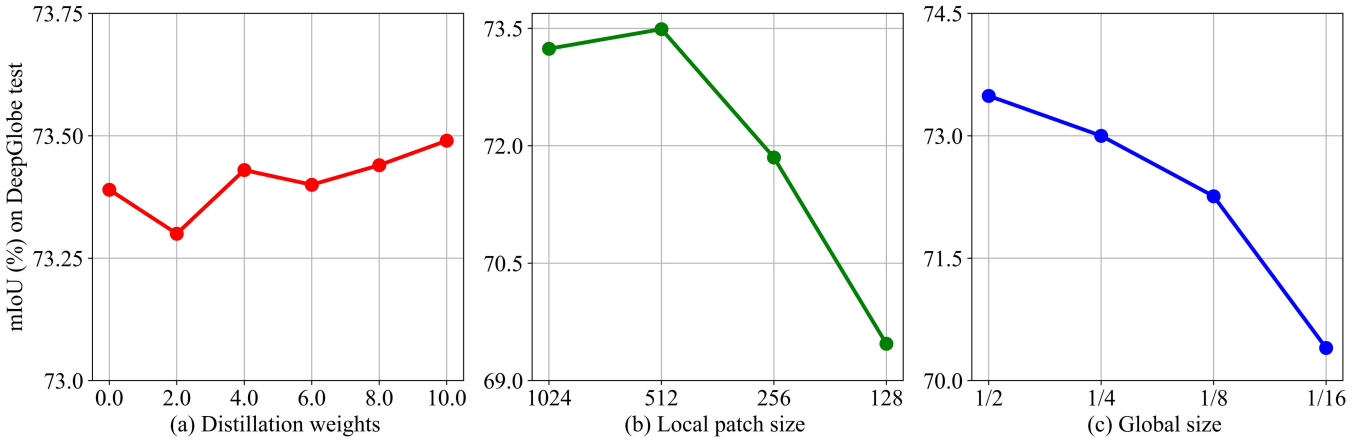


Fig. 5. Parametric sensitivity analysis. (a) Effect of different distillation weights on segmentation results. (b) Effect of different local sizes on segmentation results. (c) Effect of different global sizes on segmentation results.

TABLE II
EFFECTIVENESS ANALYSIS OF THE GGN MODULE AND LAD LOSS

	Local	Global	GGN	LAD Loss	OA	mF1	mIoU
1	✔	✘	✘	✘	87.00	81.98	71.14
2	✘	✔	✘	✘	77.09	69.54	56.10
3	✔	✔	✘	✘	87.14	82.16	71.43
4	✔	✔	✘	✔	87.23	82.33	71.60
5	✔	✔	✔	✘	88.18	83.64	73.39
6	✔	✔	✔	✔	88.20	83.70	73.49

TABLE III
COMPARISON OF OTHER FEATURE FUSION METHODS AND GGN. ADD STANDS FOR ADDITIVE FUSION, AND CAT STANDS FOR COLLOCATION

	OA	mF1	mIoU
Baseline	87.14	82.16	71.43
ADD	87.74	82.99	72.54
CAT	87.40	82.43	71.83
GGN	88.18	83.64	73.39

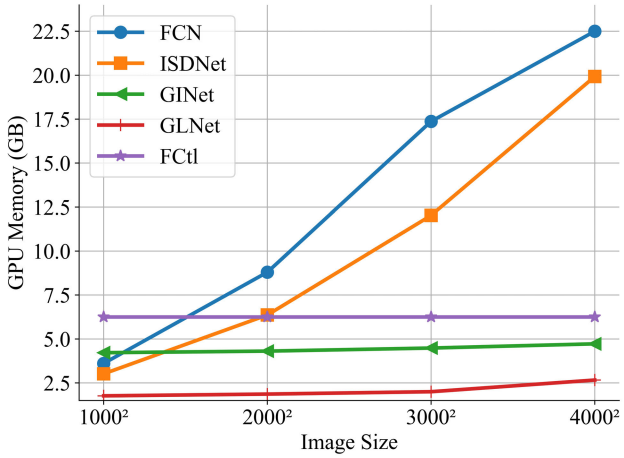


Fig. 6. Comparison of the GPU memory usage of different models. The horizontal coordinate represents the size of the image, and the vertical coordinate represents the model memory footprint.

model based on ResNet-18 with two branches, and No. 1 and No. 2 are the corresponding single branch. When we simply add global information to the local branch, the mIoU score improves from 71.14% to 71.43%. When we added our proposed GGN and LAD loss to the baseline, the mIoU score improved from 71.43% to 73.39% and 73.49%. These above results verify that our proposed improvement is effective for HR RSI segmentation.

2) Comparison With Other Fusion Methods: Our GGN module is essentially a feature fusion module that fuses the features of global branch with the features of local branch. As shown in Table III, our GCN module is able to achieve better performance compared with the traditional ADD and CAT feature fusion methods.

3) Parameter Sensitivity Analysis: We perform a sensitivity analysis on several hyperparameters in the model.

For the distillation weight λ, as shown in Fig. 5(a), when the value is taken as 10, it facilitates the alignment of spatial information between the global and local branches, and the best segmentation performance can be obtained.

For the size of local branch, as shown in Fig. 5(b), the best segmentation performance is obtained when the value is 512. Too small a value significantly degrades the segmentation performance, because it does not provide enough contextual information, and too large a value degrades the segmentation performance even if it is compensated by the global branch, which we speculate is due to feature competition with the global branch. The above ablation experiment is based on the DeepGlobe dataset, whose image size is 2448 $\times$ 2448. In our experiment, when the global downsampling rate is 1/2, the image size is 1224 $\times$ 1224, while when the local crop size is 1024 $\times$ 1024, the two sizes are very close. After passing through the series pooling layer of the network, the global feature map size is 38, and the local feature map size is 32. The GGN module clips the features corresponding to the local feature map from the global feature map. There are too many duplicate positions in the cropped feature maps each time, and

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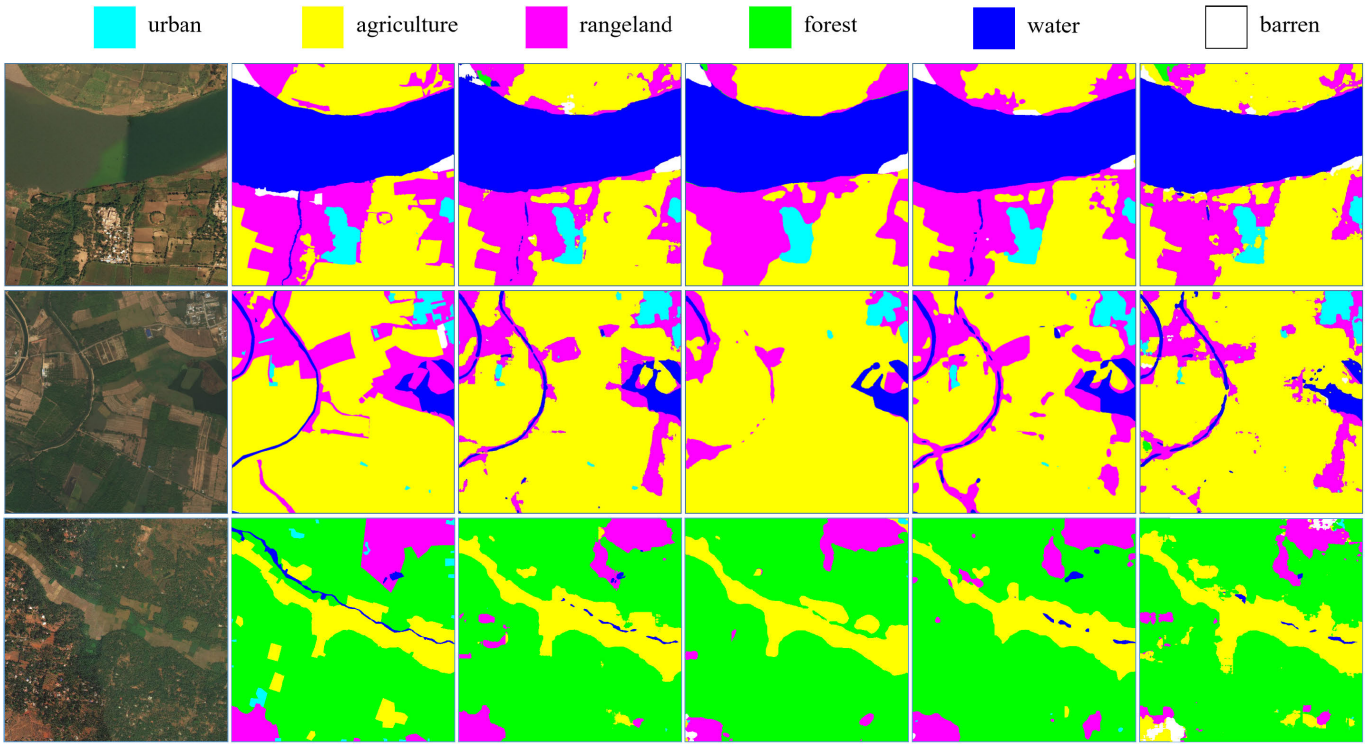


Fig. 7. We illustrate a few examples from the DeepGlobe dataset and compare them with other methods. (a) Input image. (b) Ground truth. (c) Ours. (d) GLNet. (d) FCtL. (d) Local FCN.

TABLE IV
COMPARISON OF COMPLEXITY, AND TRAINING AND TESTING TIMES OF DIFFERENT MODELS ON THE DEEPGLOBE DATASET

Method		Params.	FLOPs	Training (s)	Testing (s)
Global	Unet [6]	28.99	202.59	11593	147
	FCN-8s [4]	13.54	57.83	8968	172
	DeepLabv3+ [5]	12.3	52.95	9178	151
Local	Unet [6]	28.99	202.59	11670	536
	FCN-8s [4]	13.54	57.83	9738	271
	DeepLabv3+ [5]	12.3	52.95	9855	283
UHR	GLNet [3]	-	-	387543	864
	FCtL [7]	155.32	658.62	693857	975
	ISDNet [8]	17.78	237.44	30416	14
	Ours	50.82	255.73	33533	454

the similarity between the cropped feature maps is too high, which results in feature competition. If the local crop size is 512 × 512, the local feature map size is 16. It can cut out more diverse feature maps, reduce feature competition, and regularize the network more.

For the size of the global branch, as shown in Fig. 5(c), an oversized downsampling rate significantly degrades the segmentation performance, because it completely destroys the pixel information of the image, even with the detailed information provided by the local branch.

4) Comparison of Memory: Essentially, our algorithm is a framework for semantic segmentation of HR images and is capable of handling large-size images. As shown in Fig. 6, as the image resolution increases, the amount of memory required by our model essentially stays in a small range, while the two comparison algorithms require exponentially more memory. It can be observed that the memory consumption of our method is small and does not increase with the increase of image size, and the memory consumption is between GLNet and FCtL.

Comparison of Model Efficiency. Table IV reports the model complexity and training test time of our proposed GINet, which we use the DeepGlobe-Map dataset as the basis for our evaluation. It has an image size of 2448 × 2448, 455 training images, and 142 test images. Since our method uses the framework of combining downsampled view and cropped view, it naturally reduces the FLOPs occupancy by several times, which indicates that our method is beneficial to model training and inference under resource constraints. In addition, the introduction of the GGN module does increase the number of parameters and lengthens the training time, but the test times of our method are all in the middle of the range. The multistage hybrid framework of GLNet and FCtL takes a long time to train, because multiple models need to be trained. ISDNet, a lightweight, end-to-end approach, is capable of high-speed predictive reasoning at the expense of precision.

E. Comparison With State of the Art

In this section, we show a comparison with different methods on three publicly available datasets, including generalized segmentation networks, such as FCN [4], UNet [6], and DeepLab v3+ [5]; none of these methods are designed for HR images, and for comparison purposes, we use a cropped local view and a downsampled global view to train the model, respectively, for the following: 1) global inference, where the

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TABLE V
SEGMENTATION RESULTS FOR EACH CLASS AS WELL AS THE AVERAGE ON THE DEEPGLOBE DATASET

Method		urban	agriculture	rangeland	forest	water	barren	mIoU
Global	Unet [6]	72.59	76.70	14.56	64.44	70.94	32.07	55.22
	FCN-8s [4]	72.20	78.32	16.14	66.32	69.42	39.55	56.99
	DeepLabv3+ [5]	69.92	77.77	14.06	62.64	64.66	43.25	55.38
Local	Unet [6]	77.04	83.79	31.28	73.57	80.58	53.36	66.60
	FCN-8s [4]	77.51	85.03	33.28	76.12	81.34	54.83	68.02
	DeepLabv3+ [5]	78.45	85.99	35.45	76.74	81.36	56.65	69.11
UHR	GLNet [3]	77.16	86.77	39.69	79.34	81.24	63.36	71.26
	MagNet [35]	79.42	87.21	41.06	79.36	83.84	61.70	72.10
	MagNet-fast	75.77	84.84	33.76	75.91	80.46	58.55	68.22
	FCtL [7]	78.94	87.65	42.71	79.58	83.14	64.82	72.81
	ISDNet [8]	79.55	**88.21**	42.48	79.82	84.58	**65.20**	73.30
	Ours	**80.00**	87.94	**43.77**	**81.21**	**84.78**	63.26	**73.50**

model is trained and tested on the downsampled scale, and the final result is obtained by upsampling and 2) local inference, where the model is trained and tested on cropped images, and the final result is obtained by splicing. In addition to ordinary segmentation networks, we also compare some representative high-resolution image segmentation methods such as GLNet [3], MagNet [35], FCtL [7]. Note that since the public article lacks single-class experimental results, most of the results are obtained from our retraining inference.

1) DeepGlobe Benchmark: We report results on the DeepGlobe test set, and the comparisons are shown in Table V, with the best results highlighted in bold. We divide Table V into three sections based on different training approaches. We can observe that the local-view-based training approach achieves better performance than the global-view-based training approach, achieving an increase of about +10%, which demonstrates that the generalization performance of the model is inhibited when there is a large difference in resolution between the training and test images. Representative methods designed for HR images all achieve better performance compared with local-view-based methods. Overall, our GINet outperforms all other methods with 73.5% mIoU, improving 16.51% over the best global model FCN, 4.39% over the best local model DeepLab v3+, 2.24% over the similarly structured GLNet model, and 0.2% over the current best ISDNet model.

Fig. 7 shows that several samples from the DeepGlobe test set are shown to further visualize the results of qualitative segmentation, and the proposed GINet network achieves better segmentation results, and our GINet is able to produce semantic labeling predictions that are more consistent with the ground truth than global FCN, local DeepLab v3+, and GLNet. For categories with elongated shapes, such as "water," our segmentation results are more coherent. For large-area categories, there are fewer holes and errors in our segmentation results. In short, our method utilizes local branch to compensate for the detail portrayal ability of global branch.

2) Inria Aerial Benchmark: We experiment on the Inria Aerial dataset to further validate the proposed algorithm. We report the results on the Inria Aerial test set and a comparison with other UHR segmentation methods. As shown in Table VI, our GINet achieves significant improvements.

TABLE VI
SEGMENTATION RESULTS ON THE INRIA AERIAL DATASET

Method		mIoU
Global	Unet [6]	54.20
	FCN-8s [4]	34.00
	DeepLabv3+ [5]	45.31
Local	Unet [6]	72.49
	FCN-8s [4]	72.19
	DeepLabv3+ [5]	72.10
UHR	GLNet [3]	71.20
	CascadePSP [50]	69.40
	FCtL [7]	71.24
	ISDNet [8]	74.39
	Ours	**76.57**

Specifically, these segmentation methods, such as GLNet, FCTL, and CascadePSP [50], designed for HR images, do not outperform the generalized segmentation methods based on local views, which indicates that it becomes more difficult to capture global information as the image size increases.

Fig. 8 shows several samples on the Inria Aerial test set where the proposed GINet network achieves better segmentation results. Compared with the local FCN, for large building areas, our method is able to obtain better qualitative results because of the introduction of global information. Compared with the global DeepLab v3+, for slender ports, our method is able to segment them more completely because of the introduction of local information.

3) GID-15 Benchmark: The difficulty with the GID-15 dataset is the higher resolution and more refined class delineation compared with DeepGlobe and Inria Aerial Dataset. We report the results on the GID-15 validation set and a comparison with state-of-the-art methods. As shown in Table VII, the proposed method achieves a clear improvement, with the best results in the absolute majority of categories. We can observe that the local-view-based training approach achieves better performance than the global-view-based training approach, achieving an increase of about 20%–30%, and although downsampling yields a better global view, excessive downsampling severely affects the model's generalization ability.

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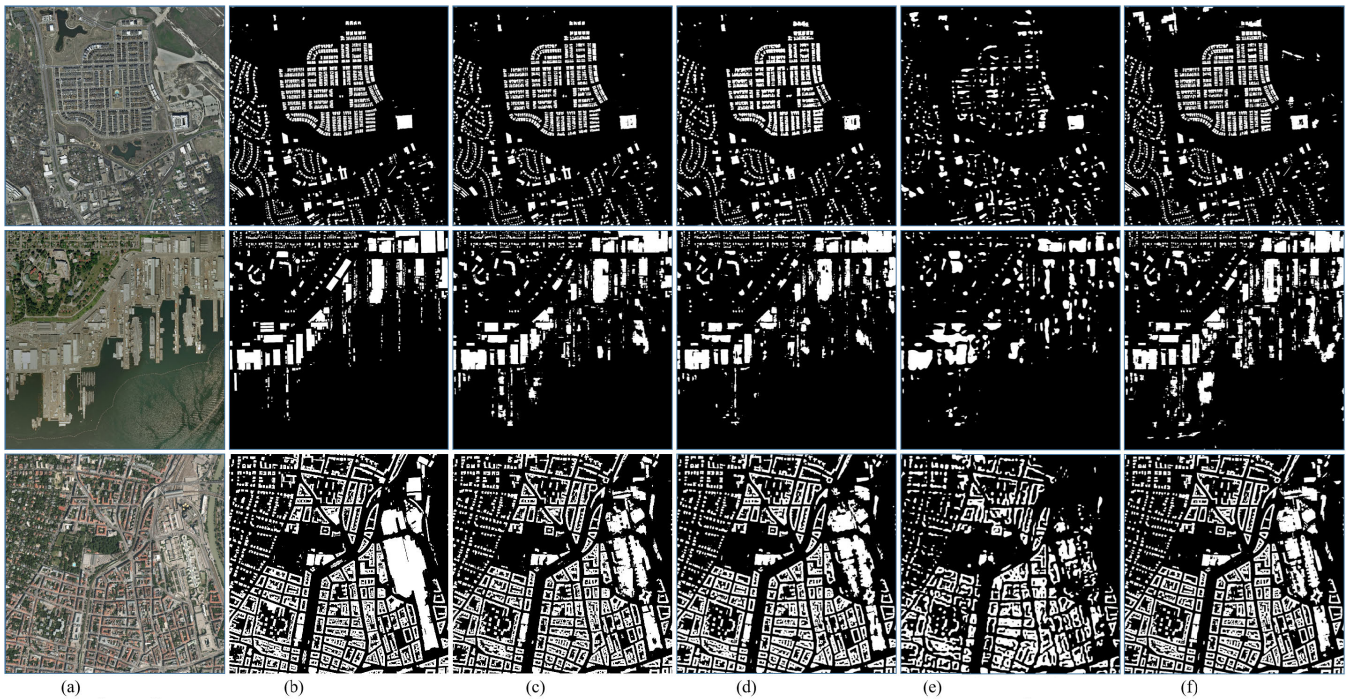


Fig. 8. We illustrate a few examples from the Inria Aerial dataset and compare them with other methods. In the figure, white and black represent buildings and nonbuildings, respectively. (a) Input image. (b) Ground truth. (c) Ours. (d) GLNet. (e) Global DeepLab v3+. (f) Local FCN.

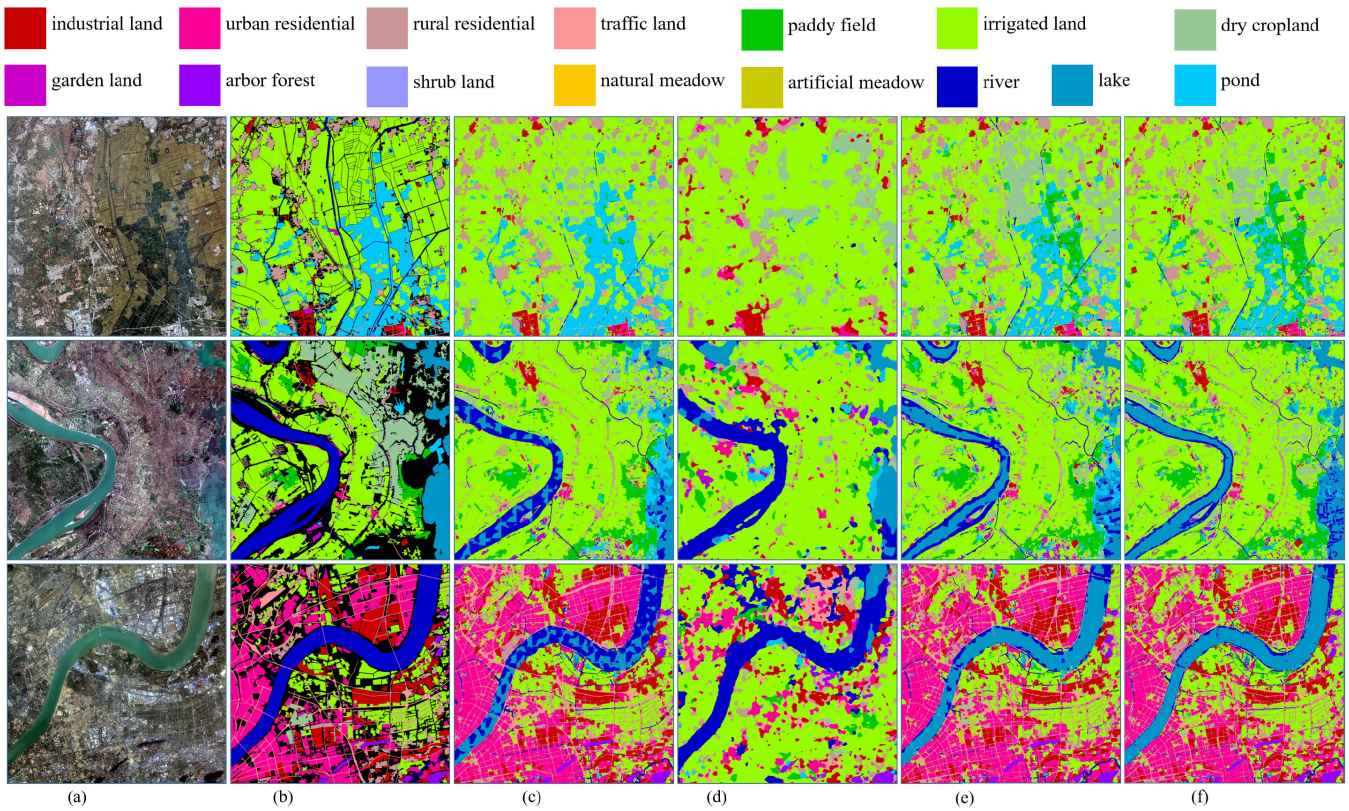


Fig. 9. Demonstration of the segmentation results of different algorithms on the GID-15 dataset. (a) Input image. (b) Ground truth. (c) Ours. (d) Global DeepLab v3+. (e) Local FCN. (f) Local UNet.

Fig. 9 shows several samples on the GID-15 validation set, where the proposed GINet network achieves better segmentation results, compared with local FCNs, and for large river areas, our method drastically reduces misclassification because of the introduction of global information. Compared with the global DeepLab v3+, for bridges over rivers, our method is able to recognize them better because of the introduction of local information.

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TABLE VII
SEGMENTATION RESULTS FOR EACH CLASS AS WELL AS THE AVERAGE ON THE GID-15 DATASET

	Method	IDL	UR	RR	TL	PF	IL	DC	GP	AW	SL	NG	AG	RV	LK	PN	mIoU
Global	Unet [6]	46.70	52.92	28.50	14.06	32.36	65.61	3.26	1.88	81.67	2.91	28.45	5.41	53.27	45.73	9.07	31.45
	FCN-8s [4]	30.77	49.10	14.12	7.23	23.14	60.05	2.22	0.64	68.87	0.27	21.65	2.32	60.65	59.86	4.89	27.05
	DeepLabv3+ [5]	39.28	39.13	18.75	6.57	24.35	56.27	0.73	0.96	65.84	2.51	30.14	1.38	53.36	46.33	3.37	25.93
Local	Unet [6]	65.20	74.27	63.01	65.03	57.64	84.83	9.17	11.61	91.90	17.66	78.50	8.10	28.67	28.78	40.98	48.36
	FCN-8s [4]	65.88	74.81	62.91	66.80	60.21	86.79	**8.66**	17.17	93.46	22.41	76.39	46.62	40.20	34.16	47.08	53.57
	DeepLabv3+ [5]	67.66	75.19	62.94	65.76	59.13	87.37	5.49	16.55	94.59	33.29	79.76	49.12	**71.30**	**51.19**	40.81	57.34
UHR	ISDNet [8]	67.87	75.11	62.42	68.12	36.72	87.05	3.92	3.97	94.26	12.66	80.87	1.49	63.68	18.12	48.19	48.30
	Ours	68.06	77.12	65.69	69.27	76.08	88.48	8.33	26.15	95.32	34.63	85.56	63.10	67.95	43.70	**62.13**	**62.10**

V. CONCLUSION

In this article, we propose a framework for semantic segmentation that is capable of handling HR images of large size, called GINet, to improve segmentation accuracy. While the classical global-view-based approach has a larger sensory field and the local-view-based approach has a strong detail portrayal capability, the framework in this article combines the advantages of both approaches. Specifically, a GGN module is proposed, where the network modulates local features based on global semantic features. In addition, we propose an LAD loss, where the feature relations of the local branches are used to guide the learning of the feature relations of the global branches. Experimental results on three land-cover classification datasets demonstrate the effectiveness and generalizability of our proposed framework. As for future work, we focus on improving the processing speed while ensuring the accuracy.

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